$\Delta B = 7.9 \times 10^{-3}$ T **A1**

4 **(a)** $F = \dfrac{Qq}{4\pi\varepsilon_0 r^2}$

$= \dfrac{(3.20\times10^{-7})^2}{4\pi(8.85\times10^{-12})(2\times0.50\sin3.0^0)^2}$ **C1**

$= 0.33616$

$= 0.34$ N (to 2sf) **A1**

(b) By considering the forces on one sphere,

$\tan\theta = \dfrac{F_E}{W}$

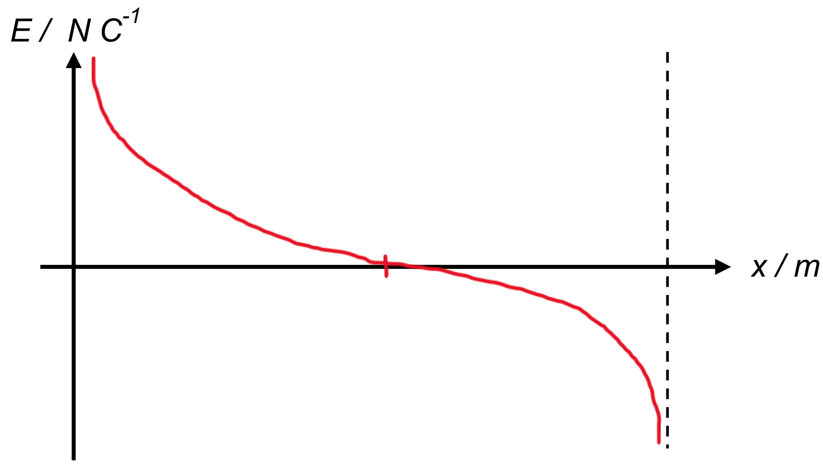
$m(9.81) = \dfrac{0.33616}{\tan 3.0^\circ}$ **C1**

$m = 0.65385$

$= 0.65$ kg (to 2sf) **A1**

(c) **(i)** The electric potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point. **B1**

(ii)

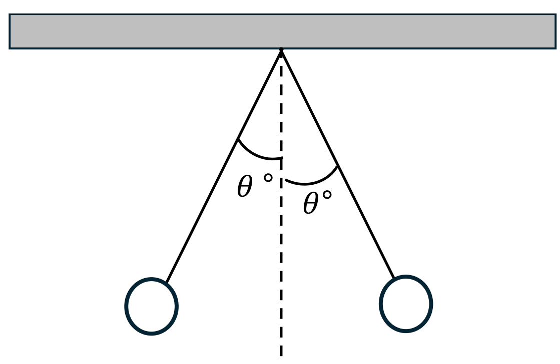


B1: Correct Shape
B1: Intercept the horizontal axis at the midway point.

(iii) Since the electric field strength E is related to the potential as $E = -\frac{dV}{dx}$, and the graph in Fig. 4.2 is E against x. **B1**

The potential difference between two points can be found by determining the area under the graph between the two points. **B1**

(d)



Since the magnitude of the electric force acting on each sphere is equal to each other (due to Newton's third law) but less than before due to the decrease in charge in one of the spheres, **B1**

each sphere will be deflected by the same, but smaller than before angle θ from the vertical. ($\theta < 3.0°$) **B1**